

Grow With Me: Exploring the Integration of Augmented Reality and Health Tracking Technologies to Promote Physical Activity

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ABSTRACT

Maintaining a healthy lifestyle requires ongoing and consistent effort. To address this challenge, a variety of technologies, such as wearable devices, have been investigated to encourage individuals to adopt healthy habits. We introduce *Grow with Me* that combines an Augmented Reality (AR) virtual pet with fitness tracking. In our application, the user's goal achievement level is linked to the health and mood of the user's virtual dog. As the user makes progress, their virtual dog learns new tricks and provides new interaction possibilities. To evaluate this solution, we conducted a mixed-methods within-subjects study with 21 participants who used both the 2D and AR versions of our application. Our results uncovered that the AR pet elicited stronger emotional responses, and participants expressed a greater willingness to use AR as a motivational tool. In this paper, we will present our study and its findings, as well as the potential of AR as a powerful tool for motivating health-related actions.

CCS CONCEPTS

• **Human-centered computing** → **User centered design; Empirical studies in HCI; User studies; Mixed / augmented reality.**

KEYWORDS

Personal Health Informatics, Augmented Reality, Emotional Design, Engagement, Persuasive Technology, Tracking Technology, Virtual Character

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1 INTRODUCTION

Many people often cite a lack of motivation as a major obstacle to engaging in exercise or adhering to a healthy lifestyle [27]. The prevailing strategies in the market, such as the Apple Watch and Fitbit, extensively leverage statistical data, encompassing charts, graphs, and analytical reports from smartwatches [29]. However, research suggests that an exclusive reliance on these quantitative methods may cause overwhelmed feelings [43] and cultivate adverse attitudes toward exercise, potentially impeding motivation to sustain fitness goals [15, 48]. Therefore, it's crucial to explore alternative methods beyond solely relying on statistical data to effectively communicate personal health information.

Research indicates that emotions play a significant role in shaping people's information processing and are a fundamental driver of human behavior [18, 30]. In the field of Human-Computer Interaction (HCI) and personal health informatics, various design strategies have been employed to harness the power of emotions to influence users' attitudes and behaviors [16, 22, 29, 38, 40, 49]. However, these design strategies have predominantly been confined to flat screens, like those on mobile phones and laptops, limiting their capacity for emotional arousal and engagement due to the lack of co-presence [25], full-body interactivity [66], vividness [66], and realism [26].

Immersive media, such as Virtual Reality (VR) and Augmented Reality (AR), has demonstrated that users tend to experience stronger emotional responses within these environments compared to a standard 2D flat screen due to the feelings of presence and immersion [36, 53]. While AR and VR have started to gain attention in the field of fitness, as demonstrated by the introduction of VR fitness games like VR Boxing and VR Sports, as well as the use of VR/AR in rehabilitation [55] and exercise posture correction [23], few studies focus on the integration of VR/AR with users' daily health data. Therefore, we decided to explore the design space of mobile AR within the context of fitness tracking, aiming to enhance user emotional attachment and subsequently promote fitness motivation.

To explore this concept that AR can enhance emotional attachment and subsequently act as a motivator for fitness, we developed a prototype entitled *Grow with Me*. This prototype integrates users' daily exercise data, such as step count, with the energy levels and mood status of a virtual dog within an AR environment. Additionally, it facilitates full-body interaction between the user and the

virtual dog, aiming to create a more immersive and engaging experience that fosters a deeper emotional bond and encourages ongoing commitment to health and fitness. In our study, we conducted a comparative analysis between mobile AR and 2D screen interfaces to assess their respective impact on emotional attachment to a virtual dog and the subsequent motivation to maintain a healthier lifestyle.

Our work offers the following contributions:

- A novel design approach aimed at promoting healthier lifestyles by integrating AR with fitness tracking, exploring AR's affordances and its potential to elicit emotional attachment and motivational drives within the context of fitness.
- A functional and interactive prototype that implements these concepts by encoding personal health data into AR with full-body user interaction.
- Findings from comparative studies that evaluate the impact of mobile AR versus 2D screens on users' emotional attachment levels and their motivation to work towards a healthier lifestyle. Based on our findings, we enumerate design strategies that can be employed in AR for fitness-tracking applications.

2 RELATED WORK

In this section, we reviewed relevant theories and past HCI research that informed our design.

2.1 Data and Emotion-Driven Health Applications

Technology has long been used to track physical activity, particularly as sensors become smaller and less expensive [21]. Currently, the mainstream health tracking applications in the market are mainly data-driven, such as using wearable devices to track users' physiological data [1, 20] and physical activity [45, 50] and then generate statistical reports [6, 32]. This data-driven approach can give users real-time feedback and comprehensive reports. However, research indicates that these quantitative methods may not effectively contribute to enhancing health outcomes because they fail to support users' emotional needs [4, 7, 15] and may contribute to fostering negative attitudes about health tracking [17]. For instance, a study utilizing Fitbit devices to monitor heart disease patients found that these individuals often felt overwhelmed by the substantial volume of data generated by self-tracking devices and applications, which resulted in challenges with comprehending and interpreting the information [42]. Hence, the data-only methods may hinder motivation for individuals who struggle to maintain a healthy lifestyle.

Some researchers and designers have recognized these issues and have attempted to address them by employing qualitative methods, such as iconic or metaphoric illustrations coupled with self-decided design [37], gamification elements [40], social sharing features [58, 64], or interactive storytelling [49]. For instance, Ubifit Garden transforms physical activity data into a virtual garden to boost users' physical activity [16] and WhoIsZuki leverages interactive storytelling to forge emotional connections between users and characters to motivate users to exercise more [49].

These design strategies have effectively evoked emotions through emotional attachment to virtual characters [40, 49], ensuring joyful experiences [37], driving a sense of responsibility towards fitness goals [64], and invoking feelings of connectedness with family or friends [57, 64]. As a result, they have proven effective in motivating individuals to engage in a healthier lifestyle. However, they have primarily been displayed on 2D flat screens, potentially limiting engagement and emotional stimulation compared to immersive 3D environments offered by emerging media [25, 26, 66].

2.2 The Virtual Pet Mechanics

The utilization of virtual pet mechanics holds potential for various interventions in the realms of study and health management, as it facilitates the establishment of emotional connections. We selected dog nurturing for our context, drawing inspiration from the Tamagotchi and other virtual pet projects. The Tamagotchi phenomenon has shown that owners often develop a deep attachment to their virtual pets and foster a genuine bond through digital "parental" behavior [13]. This concept has been widely employed in numerous research studies as a mechanism for behavioral intervention. For example, Nomi, a fitness application available on the Apple Watch, incentivizes users to feed a virtual pet based on their accumulated steps [14]. One study demonstrates that virtual pets can effectively promote physical exercise among young individuals [3, 33]. Another study has leveraged Tamagotchi-like mechanics to influence youth eating habits through mobile phone-based virtual pets [11]. These studies reveal that a bond between humans and virtual pets can be established through ongoing care-giving actions, which in turn effectively catalyze behavior change.

However, the aforementioned studies are also restricted to 2D screens, which have limitations in eliciting emotions. One study mentioned that they utilized AR along with a virtual pet to motivate users to avoid sedentary behavior [39]. However, this study was conducted earlier when AR technology primarily integrated GPS features without fully immersing users in 3D content. In addition, these studies failed to consider the potential demotivating effects of negative responses from the virtual pet when users' goal achievement is low, leading some users to discontinue their use of the application [40].

In the following sections, we enumerate how our proposed solution addresses these limitations while achieving our objective of deeply engaging users and providing lasting motivation for a sustained healthier lifestyle.

2.3 AR for Enhancing Emotional Experience

Compared to 2D flat screen media, immersive technologies like VR, AR, or XR have the potential to enhance emotional experience due to the vividness [66], full-body interactivity [66], realism [26] and the sense of presence [25, 44, 65], which consequently increase the persuasiveness for attitude and behavior change [31, 60].

Numerous studies have illustrated the capacity of AR experiences to not only enhance emotional engagement but also to influence users' behaviors and attitudes [8, 12, 47, 54]. For instance, Jung et al. conducted a study comparing the effectiveness of 2D and AR mediums in a smoking cessation campaign. Their findings revealed that the AR group not only experienced more intense negative emotions

but also exhibited a stronger intention to quit smoking, compared to the 2D group [34]. Furthermore, in the field of museum design, researchers demonstrate that AR can significantly increase enjoyment and promote empathetic behavior [24, 59]. However, investigations into how the affordances of AR can be utilized to cultivate a sustained healthier lifestyle remain significantly under-explored. Although there are studies illustrating how Pokémon Go promotes physical activity [35], it is crucial to recognize that it remains a game not explicitly designed for fitness purposes. This oversight underscores a significant area for further research, emphasizing the potential of integrating AR with health-tracking technology to make a meaningful impact on long-term health and wellness.

3 DESIGN PROCESS AND SYSTEM RATIONALE

We developed the prototype of *Grow with Me* to explore the application of AR in health tracking and its potential to increase emotional engagement, thereby motivating users to adapt healthier lifestyles. We chose mobile AR over VR because of its accessibility [52].

The core concept in our design involves linking the user's exercise data, such as step counts, to the energy level and mood of an AR dog. The behavior, health status, and mood of the AR dog respond dynamically to the user's goal achievement levels. Specifically, as the user's goal achievement levels improve, the virtual dog becomes healthier, happier, and more receptive to learning additional tricks. We employed the approach of Research through Design (RtD) [67] to examine the problem space, generate pertinent concepts, and empirically test these on an experimental artifact.

This section details the design process and rationale behind *Grow with Me*, focusing on two key aspects: 1) The simulated scenario design to determine the correlation between the dog's behavior and the user's achievement level. 2) The AR interaction design to develop the AR experience and user interaction.

All procedures involved in this project have received approval from the Institutional Review Board (IRB).

3.1 The Simulated Scenario Design

To elucidate the relationship between dog behaviors and people's emotional reactions, and thereby inform the user scenario design, we initiated an online survey (see Appendix C). This survey received 45 responses. Given that the virtual pet we were designing for was a dog and hence the intended user base would be individuals who enjoy the idea of caring for a pet dog, we included a few screening questions to ensure that our survey responders were either dog lovers or dog owners. The data helped identify the dog behaviors and the varying degrees of user reactions they generated (see Appendix E). These observations guided the design of our AR dog, ensuring that its behavior would resonate with and impact users.

The insights gathered from our online survey were instrumental in developing the "Five Scenarios of Goal Achievement" table (see Appendix A), which delineates the behaviors of the AR dog corresponding to various stages of users' goal completion. In designing the goal achievement levels, we drew upon Locke and Latham's Theory of Goal Setting [41], and established 25% intervals as our incremental progress milestones by referring to WholsZuki's milestone design for incremental progress [49]. This approach was also

shaped by considerations of developmental constraints and the necessity for simplicity during the prototype stage.

Then, we adapted the intervention design from the Fish'n'Step project [40], incorporating several modifications. Similar to the original project, we employ both positive and negative reinforcement to establish a link between user activity and the dog's responses. More specifically, the dog's reactions are calibrated to correspond with the user's goal achievement levels, ranging from negative through neutral to positive as the user progresses from low to high achievement. However, to mitigate the potential demotivating effects of negative responses [40], we introduced interactive features that allow users to comfort their dog when their goal progress is low. This addition aims to encourage continued engagement and strengthen the emotional bond between the user and the virtual dog.

3.2 Interface and AR Design

3.2.1 Interface Design. Building upon the "Five Scenarios of Goal Achievement," our prototype is designed with five distinct scenarios, each representing different levels of user goal achievement. The interface design draws inspiration from established fitness applications and research such as Apple Health, Fitbit, and WhoIsZuki [49], incorporating several essential components: the User Profile, Goal Achievement and Weekly Progress, with the addition of a Dog Profile (see Fig 1).



Figure 1: Interface Design: a) User Profile b) Goal Achievement and Weekly Progress c) Dog Profile d) Dog in AR Mode

One of the key components in our interface design is the Dog Profile, which contains details about the virtual dog, such as age, height, weight, health, and mood level, as well as the tricks it has learned. Notably, the dog's health and mood levels are dynamically linked to the user's goal achievement level, creating an interactive and responsive experience. Additionally, under the Dog Profile, users can access the AR dog feature, which will be discussed in more detail in the following section.

3.2.2 AR Interactive Design. The AR dog's behavior dynamically responds to the users' achievement levels. Upon accessing the dog page in AR mode, users can view their virtual dog present in their immediate environment and encounter different scenarios based on their progress:

- **Level-1 (0%-25%):** At this initial stage, users will see the dog lying on the ground listlessly, with a bowl of dog food in

front of it. Users can attempt to drag the food bowl towards the dog using their fingers. However, as the bowl approaches, the dog will refuse to eat and emit a sad whimper, reflecting the low level of user achievement (see Fig 2a).

- Level-2 (25%-50%): With some progress, the dog gains a bit of energy from the user's physical activities but remains in a bad mood, expressing its dissatisfaction by angrily barking at the user. If the user faces the dog, it continues to bark, but it will stop when the user turns away (see Fig 2b).
- Level-3 (50%-75%): At this intermediate stage, the dog will sit on command but show reluctance to engage in play, such as refusing to fetch a ball thrown by the user (see Fig 2c).
- Level-4 (75%-100%): As the user nears their goal, the dog becomes visibly happier, demonstrating affectionate behaviors such as nuzzling. When the user leans down and touches its chin, the dog responds positively by wagging its tail and nuzzling back (see Fig 2d).
- Level-5 (100%+): Once the user reaches or surpasses their goal, the dog is fully energetic and eager to play, jumping towards the user and adjusting itself to the face the user as they move around (see Fig 2e).

These varying behaviors are designed to visually and emotionally reflect the user's progress, providing a tangible and interactive representation of their health and fitness achievements.

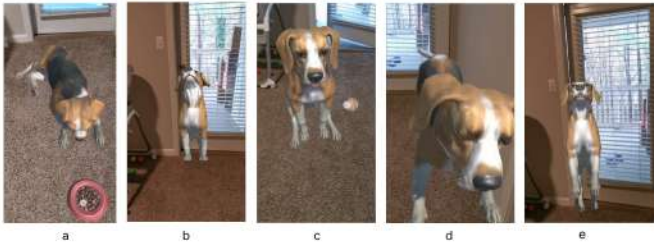


Figure 2: AR Interactive Design: a) Level-1 Refuse to eat, b) Level-2 Bark, c) Level-3 Refuse to play, d) Level-4 Wag and nuzzle, e) Level-5 Play and jump

4 STUDY DESIGN

To better understand whether the integration of AR with fitness-tracking technology can effectively elicit emotional attachment and subsequently motivate physical activity, we conducted a comparative study between AR and traditional 2D interfaces. To obtain a comprehensive understanding, we adopted a mixed-methods approach, combining quantitative and qualitative methods to evaluate the user experience with both versions of our application. Quantitative measures were used to assess changes in emotional attachment, while qualitative insights were gathered from semi-structured interviews, providing a deeper understanding of user perspectives.

4.1 Study Design and Procedure

We recruited 21 participants (13 females, 8 males, 18–41, median age 22). A within-subjects study compared participant experiences in AR and 2D screen conditions. To reduce carryover and ordering

effects, individuals were assigned at random to start one condition before switching. Experiments were conducted on Google Pixel 5 with audio recording capturing user interactions. The entire experiment lasted approximately 30 to 40 minutes, and participants received a \$10 Amazon gift card as compensation.

Participants engaged in tasks involving five scenarios representing various levels of health goal achievement and the virtual dog's corresponding behaviors. Researchers introduced participants to the interface's basic features. Since this was a prototype for user feedback about AR in fitness, a Wizard of Oz [19] version was employed. In this setup, researchers explained and switched between scenarios during the study without collecting real fitness data from users. Instead, dummy data was employed to represent users' goal achievement levels. For example, the researcher might say, "Imagine you've only completed 20% of your goal. Please check on your dog" (for the Wizard of Oz script, please refer to the Appendix F). Participants were not timed and were encouraged to observe and interact with the virtual dog for as long as they wanted.

The distinction between the 2D and AR versions lies solely in the presentation of the virtual dog. In the 2D condition, participants viewed an animated dog on their mobile screen accompanied by audio, but without any interactive features. Conversely, in the AR condition, the virtual dog appeared within the participant's physical environment, allowing for interaction. For instance, in level 1, assuming the user had only completed 20% of their goal, in the 2D condition, the user would see a dog depicted on the screen, lying listlessly on the ground while emitting whimpering sounds. In contrast, in the AR condition, the user would observe a virtual dog lying listlessly on the ground in front of them, with the ability to view it from different angles within their physical space. As they attempted to drag the food bowl to the dog, it would refuse to eat and continue whimpering (see Appendix D).

Then, participants completed a card sorting activity, ranking cards that depicted the dog's behaviors based on the strength of the emotional connection they felt towards each behavior while also providing explanations. They also filled out the Emotional Attachment Scale, discussed further in the Results section, measuring their empathy towards the virtual dog. After completing one condition, they transitioned to the other and the same study flow was repeated.

After trying both conditions, participants were involved in semi-structured interviews. These interviews compared the AR and 2D screen conditions, covering user preferences, predicted impact on exercise levels and behaviors, potential challenges, expectations, and suggestions for application improvement.

4.2 Data Analysis

We analyzed the quantitative data in the Emotional Attachment Scale by an independent samples t-test using JASP software [62]. The Shapiro-Wilk test indicated that the data did not significantly depart from a normal distribution (2D version: $W(20) = .95$, $p = .419$; AR version: $W(20) = .97$, $p = .783$). This suggests that the assumption of normality was met for both conditions, allowing for the t-test to be conducted. For the qualitative interview data, we conducted open coding with thematic analysis [9] by using ATLAS.ti [5].

5 RESULTS

5.1 Emotional Attachment to the Virtual Dog

In our study, we employed the Emotional Attachment Scale to measure and compare users' emotional attachment in both 2D and AR conditions. This scale was adapted from the Narrative Engagement Scale [10], with a particular emphasis on the components of empathy and sympathy. We selected this scale due to its ability to measure the level of emotional attachment with the character, as it has been successfully applied in similar studies, such as the one conducted by Whoiszuki [49]. The sample question, such as "At important moments, I could feel the emotions the dog felt," was utilized to gauge the participants' emotional connection with the character (For the Emotional Attachment Scale, please check Appendix B).

The t-test results indicate a statistically significant difference in emotional attachment between the 2D condition (mean = 6.73, standard deviation = 1.07) and the AR condition (mean = 7.0, standard deviation = 1.23), with a p-value of 0.048, cohen's d = 0.2342. Quantitative analysis reveals that emotional attachment is higher in AR experiences compared to 2D. Moreover, the calculated p-value confirms the statistical significance of this difference.

In addition, we used card sorting to determine which kind of virtual dog behaviors can create deeper emotional attachment, providing insight for future design. Following each condition, participants ranked the dog's behaviors in the five scenarios on a scale of 1 (weakest) to 5 (strongest). For example, if a participant felt the strongest emotion for the behavior depicted in scenario Level 3, it was given a rating of 5. Subsequently, we calculated the average rating for each behavior observed in the virtual dog. In the AR condition, users said interaction was the key to engagement and therefore Level 5, where the dog jumps facing the user and adjusts its direction as the user moves around, received the highest average score due to the full-body interaction. They also named audio and animation as the second and third important design aspects enhancing the experience.

Furthermore, when comparing 2D interfaces with AR, participants noted that AR tends to foster a stronger emotional connection. "P8: I felt obviously more emotion towards AR."

5.2 AR's Advantages and Disadvantages in Health Tracking

In terms of preference for AR or 2D, 17 of 21 participants preferred AR, two chose 2D, and two chose either. The reasons for picking AR are listed below.

5.2.1 The interaction in AR could improve user engagement in health-tracking activities. A significant majority of the participants (17 out of 21) expressed a preference for AR over 2D screens. This preference was attributed to AR's enhanced interactivity and the heightened sense of presence it provides. Regarding motivation, 15 out of 21 participants favored AR, citing a stronger emotional connection with the AR dog as a key factor. "P6: They are one because since I feel a stronger connection to the AR version of the dog, I would be more motivated to complete my goal with that version than the 2D." Additionally, participants exhibited curiosity about the novel interaction possibilities offered by AR, which in turn would

motivate them to engage in more exercise. This increased activity was driven by their desire to unlock new ways of interacting with the virtual dog. "P31: I think just having that extra level of interaction between you and the dog where not only does your level of exercise affect how they're feeling, but it affects how much they want to interact with you. I thought that was really cool."

5.2.2 The sense of presence in AR could enhance reality, build empathy and motivate users to exercise more. We found that AR could create a sense of presence, allowing users to perceive virtual characters as existing within their own environment and standing alongside them. This experience of presence facilitated a deeper understanding of the virtual pet, imparting a more realistic and tangible feeling to the interaction. "P31: I think I prefer the AR because it's like the dog is actually in the same environment as me and it's actually like in my space and it almost simulates like actually having a pet to care for like in your home environment."

5.2.3 The limitations of AR in health tracking. Two major points of contention were shared by the participants who preferred the 2D version. First, they were concerned that AR would require more physical space to display the content, making it inconvenient to open an application "on-the-go" and adding constraints on time and place of use. Second, several participants expressed a preference for the 2D version due to the presentation of health statistics. They found it more convenient as they didn't need to switch between the AR environment and their mobile screen to view fitness data. Consequently, when designing the AR version, it's crucial for designers to consider integrating these fitness metrics seamlessly into the AR interface.

6 DISCUSSION

In this study, we investigated the design space of using AR in health tracking systems to generate emotional attachment and promote exercise. Our research addresses the HCI community's call to explore new approaches to generate qualitative experiences using personal data [61, 63]. In this section, we discuss how AR designs can contribute to existing digital health interventions and spark a discussion about AR in health-tracking applications.

6.1 AR to foster emotional attachment in health-tracking applications

XR is known as an "empathy machine [46]," owing to its unique affordance of presence and perspective-taking [2, 28]. In our study, we harnessed these characteristics of AR by adopting a Tamagotchi-like approach by connecting users' health-related data to the care and nurturing of a virtual pet. Our findings indicated that the sense of presence afforded by AR, which elicited the participants' feeling that the AR dog was sharing their environment, made it easier for our participants to establish an emotional bond and commitment to the virtual pet. This emotional connection served as a potential motivator for our participants to engage in more physical activity, driven by the desire to improve the health and happiness of their AR dog.

6.2 AR to enhance engagement for health-tracking applications

Grow with Me integrates both positive and negative reinforcements similar to the previous Fish'n'Steps research study [40]. The Fish'n'Steps experiment, however, found that a negative reinforcement, seeing a depressed fish, decreased some users' engagement as users do not want to see the depressed fish [40]. To address this potential issue, we allowed our users to comfort their virtual dog when their goal progress is low. Adding this additional intervention in response to the negative reinforcement has shown the potential to continue user engagement despite having the negative reinforcement in our design. Our observations and interview results also revealed that our participants were motivated to interact with the dog, regardless of their low achievement level.

6.3 Design challenges for AR in health-tracking applications

While AR offers significant potential advantages in emotional elicitation and user engagement [44, 56], our findings suggest design challenges that must be addressed. First, a connection between health data and AR content requires greater clarity. Designers should focus on more effectively integrating statistical data within the AR content. For example, we may need to add a progress bar along with AR to indicate goal completion. Moreover, the level of interactivity in AR is an important factor for sustaining user engagement. Further enhancing interactivity, such as enabling the virtual dog to accompany users on walks or making it respond to voice commands, could significantly enrich the user experience. Addressing these challenges can advance the potential of using AR in health and fitness applications.

7 LIMITATIONS AND FUTURE WORK

Our study has provided valuable insights into the potential of AR in eliciting emotional attachment and motivating physical exercise. However, several limitations and areas for future research deserve attention.

First, prototype development issues. The prototype utilized in our study serves as a foundational exploration into the comparative effectiveness of AR versus 2D pet interfaces in fostering emotional attachment and exercise motivation. However, it is essential to acknowledge that the prototype is not fully functional, as it does not integrate real fitness data. Instead, dummy fitness data was employed to represent user achievement levels. Future iterations will involve connecting the application with real fitness data to enhance its functionality and authenticity.

Second, methodological considerations. Our experiment employed the Wizard of Oz method, enabling researchers to manipulate user scenarios and fitness achievement levels dynamically. While this approach offered valuable insights, it is not without limitations. The inability to simulate all aspects of user interaction poses a constraint on the generalizability of our findings. Future research could explore alternative methodologies to address this limitation more comprehensively.

Third, a need for field and long-term studies. While participants in our study expressed intentions regarding their future exercise behaviors based on interactions with the AR pet, there remains

a notable gap between stated intentions and actual behavior. To bridge this divide, future research will incorporate field studies and long-term evaluations using fully functional applications. These studies will provide a more nuanced understanding of how AR interfaces influence sustained exercise engagement over time.

Fourth, technical challenges. Being an early prototype, some unforeseen technical issues arose during testing that impacted overall user experience. In some instances, rendering limitations positioned the AR dog above ground level, seemingly 'floating' in the air, causing shortcomings in the realism of the dog model. Addressing these technical challenges is crucial for optimizing user engagement and immersion. Future versions of the application will prioritize improvements in model quality and AR performance to deliver a more seamless user experience.

In conclusion, while our study lays a foundation for understanding the potential of AR in promoting emotional attachment and exercise motivation, it is imperative to address the identified limitations and pursue avenues for future research. By iteratively refining prototypes, adopting robust methodologies, conducting long-term studies, and addressing technical challenges, we can advance the development and application of AR technologies in promoting health and well-being effectively.

8 CONCLUSION

In this paper, we present *Grow with Me*, a mobile AR prototype that integrates personal fitness data and AR to elicit emotions and increase user engagement. Our within-subjects study with 21 participants who used both the 2D and AR versions of our application revealed that AR is effective at triggering strong emotional responses and has higher potential as a motivational tool than a traditional 2D alternative. Our findings shed light on several design strategies and challenges associated with the usage of AR in health-tracking applications. For future studies, we intend to develop a fully functional application incorporating real fitness tracking features. We plan to conduct a long-term field study involving a larger participant pool to evaluate how this design performs in real-world settings over a prolonged period.

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A FIVE SCENARIOS OF GOAL ACHIEVEMENT

Goal Completion	Virtual Dog Behavior	User Interaction
Level-1 (0-25%)	Dog refuses to eat and whimpers (+ audio)	Users can select and drag a food bowl to feed their dog, but, the dog refuses to eat and whimpers.
Level-2 (25%-50%)	Dog barks incessantly (+ audio)	Dog barks when the user is facing the dog and stops once the user turns away from the dog.
Level-3 (50%-75%)	Dog sits on command but refuses to play with user	User can select and drag a ball on the ground to play with the dog, but the dog remains still.
Level-4 (75%-100%)	Dog wags its tail and shows a nuzzling motion	User needs to lean down and can touch/stroke the dog's chin and in response the dog wags its tail and nuzzles.
Level-5 (100%+)	Dog shows an eagerness to play with the user and jumps	The dog jumps facing the user and adjusts its direction as the user turns around.

Table 1: Five Scenarios of Goal Achievement

B QUESTIONNAIRES

Emotional Attachment Scale

C FORMATIVE SURVEY

Formative Survey

D 2D V.S. AR CONDITION



Figure 3: Left: Dog in 2D condition; Right: Dog in AR condition

E SURVEY RESULT

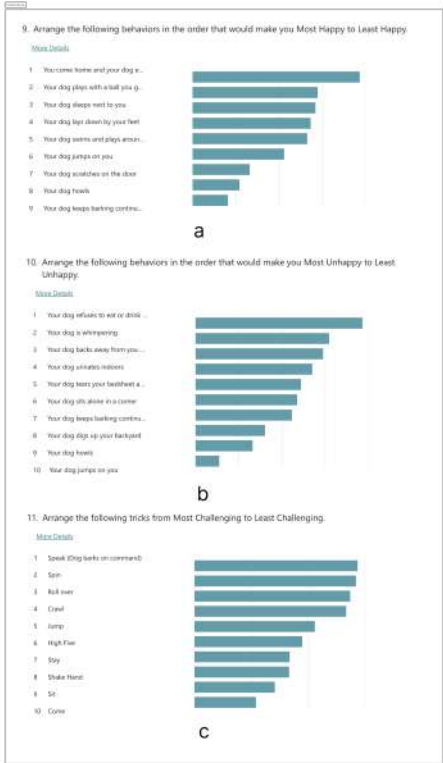


Figure 4: a) Emotional gradation of positive behaviors (Most Happy to Least Happy) b) Emotional gradation of negative behaviors (Most Unhappy to Least Unhappy) c) Reward gradation (Most Challenging to Least Challenging tricks)

F WIZARD OF OZ SCRIPT

Wizard of Oz Full Script